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Energy Policy

1. Policy Statement

Our organization is committed to improving energy performance, reducing energy consumption, and minimizing the environmental impact of our operations. We aim to integrate energy efficiency into all business processes and decision-making.

2. Scope

This policy applies to all departments, employees, facilities, equipment, and operations managed by the organization.

3. Objectives

- Reduce overall energy consumption and improve energy efficiency.
- Comply with all applicable legal and regulatory energy requirements.
- Continuously improve energy performance through systematic energy management.
- Promote use of renewable and clean energy sources where feasible.
- Increase awareness and engagement of employees in energy conservation efforts.

4. Commitments

- Establish measurable energy objectives and targets.
- Provide adequate resources to implement energy management programs.
- Conduct regular energy audits and implement identified improvements.
- Monitor, measure, and report energy performance.
- Train and motivate employees on best energy practices.

5. Roles & Responsibilities

Top Management: Approve and review energy policy; allocate resources.

Energy Manager: Lead the energy management system; report performance.

All Employees: Follow energy-saving practices; participate in improvement initiatives.

6. Review and Update

This policy will be reviewed annually and updated as required to reflect changes in operations, regulations, and energy performance goals.

APPROVED BY : Sandeep. G. Parvatikar DATE : 10/01/2025

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Energy Management Plan

1. Purpose

To optimize energy usage across the organization, reduce waste, and support sustainable manufacturing practices.

2. Scope

Applies to all equipment and activities associated with:
- CNC machining (turning, milling, grinding)
- Ancillary equipment (air compressors, pumps, HVAC)
- General lighting, desktops, laptops and facility energy usage

3. Energy Sources Identified

- Electrical Power: CNC machines, lighting, HVAC, pumps, and ancillary equipment
- Compressed Air: Tool changes, work holding, cleaning
- Hydraulics: Powering clamps and certain equipment
- Gas / Steam (if applicable): Surface treatments, washing stations
- Renewables (if applicable): Solar PV installations

4. Targets for Energy Efficiency

Short-Term (1–3 months): Minimize idle equipment running times, Switch off equipment when not in use, Maintain air compressors and fix air leaks.

Medium-Term (3–12 months): Reduce energy usage per part by 5–10%; Install LED lighting and sensor controls; Optimize machining speeds and feeds for energy efficiency.

Long-Term (1–3 years): Invest in high-efficiency equipment and servo motors; Install solar PV or other renewable sources; Achieve an overall energy reduction of 10–20%.

5. Action Plan

Action Item | Owner | Timeline | Status

- Audit equipment energy usage | Maintenance / Engineering | Month 1 |
- Train staff in energy conservation practices | Supervisor / Safety Team | Month 1–2 |
- Install energy-efficient lighting and controls | Maintenance | Month 3–6 |
- Establish energy usage monitoring and review | Engineering / Management | Ongoing |
- Evaluate equipment upgrade options | Engineering / Management | Month 12 |
- Install solar PV or other renewable source | Management / Engineering | Month 18–36 |

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6. Monitoring & Verification

Maintain an Energy Log for equipment and departments. Perform quarterly reviews to assess progress towards targets. Benchmark results against prior periods and industry best practices.

7. Continual Improvement

Incorporate energy efficiency into equipment procurement policies. Regularly review and adjust energy targets. Maintain open communication with staff about energy conservation measures.

Approval

Approved By: Sandeep G. Parvatikar

Position: Director

Date: 10/01/2025

AVG. SOLAR POWER PRODUCTION/MONTHLY - 8000Kwh (TARGET 5% INCREASE YEARLY)

SL.NO	MONTH	POWER CONSUMPTION IN Kwh	REMARKS
1	Jul-24	9,136 kWh	MORE
2	Aug-24	7,498 kWh	LESS
3	Sep-24	8,572 kWh	MORE
4	Oct-24	9,110 kWh	MORE
5	Nov-24	8,060 kWh	MORE
6	Dec-24	6,967 kWh	LESS
7	Jan-25	7,456 kWh	LESS
8	Feb-25	8,815 kWh	MORE
9	Mar-25	8,934 kWh	MORE
10	Apr-25	10,866 kWh	MORE
11	May-25	10,301 kWh	MORE
12	Jun-25	9,551 kWh	MORE

REVIEWED BY: Sandeep G. Parvatikar

DATE: 30/06/2025

Remarks: In the month of Aug 24, Dec 25 & Jan 25, the solar power production is less then the avg. due to weather condition like rain, cyclone effect. Hence corrective action is not required

12.2

HIGHEST POWER CONSUMPTION - 22Kwh (TARGET 10% REDUCED YEARLY)

SL.NO	MONTH	POWER CONSUMPTION IN Kwh	REMARKS
1	Jul-24	13,683 kWh	LESS
2	Aug-24	16,360 kWh	LESS
3	Sep-24	16,169 kWh	LESS
4	Oct-24	14,929 kWh	LESS
5	Nov-24	15,999 kWh	LESS
6	Dec-24	17,414 kWh	LESS
7	Jan-25	20,645 kWh	LESS
8	Feb-25	17,100 kWh	LESS
9	Mar-25	17,683 kWh	LESS
10	Apr-25	19,061 kWh	LESS
11	May-25	19,783 kWh	LESS
12	Jun-25	16,186 kWh	LESS

REVIEWED BY : Sandeep G. Parvatikar

DATE : 30/06/2025

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UNNATHI CNC TECHNOLOGIES PVT LTD | UCT/R/TRG-03/R2

TRAINING & EFFECTIVENESS RECORD

Title of Training program: Energy Management

Date of Training : 08/03/2025

Faculty : Suresh Kumar .S

Place of Training : Unnathi CNC Technologies Pvt Ltd

Brief description of training: Purpose, scope, energy sources, Targets for energy efficiency, Action plan, monitoring & verification, Continual improvement

SI.No.	Name	Department	Signature	Effectiveness of Training			Effectiveness measured by		
				G	S	R	A	B	C
1	Rajasekharak.N	Marketing		✓					✓
2	Naveen DB	Design		✓					✓
3	Nandan Kumar	Quality		✓					✓
4	Sreedhar KM	Turning		✓					✓
5	prashanth GPD	Milling		✓					✓
6	M. Arul	Development		✓					✓
7	Murali	Manager		✓					✓
8	Nadhiya.S	Sales			✓				✓
9	Vilas M.B	Maintenance		✓					✓
10	Sumithra	Purchase			✓				✓
11	Anil Thakur	Forging			✓				✓
12	Jyothi PRASADP	Grinding			✓				✓
13	Niranjana .C	Milling			✓				✓
14	Ravi Kumar	WireCut			✓				✓

Legend:		
	A = Personnel Discussion,	G=Good
	B = Demonstration/Test	S=Satisfactory
	C = On-the-job Assessment	R=Retrain

	Effectiveness assessed by	Evidence of Training	
Sign		Training in-charge	
Name	Sandeep G Parvatikar	MR	
Date	28/03/2025	Quality in-charge	
		Production in-charge	Murali
		Director	

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Energy Management Audit Checklist

General Information

Checklist Item	Status (Yes/No)	Remarks
Energy policy documented and approved?	Yes	
Energy management team established?	Yes	
Baseline energy consumption recorded?	Yes	
Energy goals and KPIs defined?	Yes	
Compliance with legal/regulatory requirements?	Yes	

Energy Data Monitoring

Checklist Item	Status (Yes/No)	Remarks
Energy meters installed at major loads?	Yes	
Real-time monitoring system available?	Yes	
Regular energy performance reports prepared?	Yes	
Power factor and load factor tracked?	Yes	

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Lighting Systems

Checklist Item	Status (Yes/No)	Remarks
LED lights installed in all areas?	Yes	
Motion sensors/timers used where applicable?	NO	'OI'
Daylight integration optimized?	Yes	
Lux levels checked and within norms?	Yes	

HVAC Systems

Checklist Item	Status (Yes/No)	Remarks
Thermostat set at energy-efficient range?	No	OI
Regular maintenance of HVAC units done?	Yes	
VFDs installed on fans/pumps where feasible?	Yes	
Cooling towers and chillers operating efficiently?	Not Applicable	

Motors, Pumps & Compressors

Checklist Item	Status (Yes/No)	Remarks
Energy-efficient (IE3/IE4) motors installed?	Yes	
Air leakages in compressor lines checked?	Yes	
Preventive maintenance program in place?	Yes	

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Process Energy Use

Checklist Item	Status (Yes/No)	Remarks
Heat recovery systems used (e.g., economizers)?	NO	not applicable
Insulation of steam/hot water lines intact?	NO	not applicable
Use of renewable energy considered?	Yes	Solar panels installed
Load scheduling during off-peak hours?	Yes	

Building Envelope

Checklist Item	Status (Yes/No)	Remarks
Roof and wall insulation adequate?	Yes	
Windows and doors sealed properly?	Yes	
Solar heat gain minimized through design?	Yes	

Water & Steam Management

Checklist Item	Status (Yes/No)	Remarks
Condensate recovery system operational?	NO	Not applicable
Water heating optimized for demand?	No	Not applicable.
Water leakages identified and fixed?	Yes	

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Employee Awareness & Training

Checklist Item	Status (Yes/No)	Remarks
Energy training provided to staff?	Yes	
Suggestions from employees encouraged?	Yes	
Energy-saving posters/signage displayed?	Yes	

Energy Performance Improvement

Checklist Item	Status (Yes/No)	Remarks
Energy audits conducted periodically?	Yes	
Energy-saving initiatives implemented?	Yes	
Action plan for continuous improvement available?	Yes	

Audited by : Nadhiya Date : 16/04/2025

Reviewed by : Suresh Kumar. S Date : 16/04/2025